Kartikeya Goel

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Education

Stanford University Expected Jun 2029
B.S. Computer Science; Minor in Statistics — GPA: 3.66 *Stanford, CA*

- **Coursework:** CS 107 (Computer Organization & Systems: C, x86-64, heap allocator), CS 106B (Data Structures & Algorithms), MATH 104 (Matrix Theory), MATH 51 (Linear Algebra & Multivariable Calculus), STATS 117 (Probability)
- **Fall 2026:** CS 103, CS 109, CS 161, CS 229

Experience

Apple Jun 2026 – Sep 2026
Software Engineering Intern *Sunnyvale, CA*

- Shipped a localization tool replacing a manual per-screen process across 37 locales, owned from design through deployment; used by 200+ designers and producers.
- Built internal GenAI tooling embedded in production screen workflows, wrapping model calls in deterministic pre- and post-processing so every output is schema-validated before entering the pipeline.

Sago Jan 2026 – Present
Software Engineering Intern

- Own the brief-generation service end to end: ingests company and market data, drafts structured investment briefs, and serves them as reports queried live in deal meetings.
- Built the retrieval and citation layer that resolves each generated claim to a source document.

MARi Jun 2025 – Aug 2025
Software Engineering Intern

- Parallelized processing of multi-GB inputs across 23 worker processes, reducing a 17-hour job to 30 minutes.
- Built the eval harness for an automated decisioning pipeline; found a prompt-injection failure mode and raised the pass rate from 5/23 to 22/23 cases.

IQ Spectra Inc. Mar 2023 – Apr 2025
Software Engineering Intern

- Traced and fixed a memory leak in the C/C++ parser using Valgrind and GDB.
- Re-keyed the HTML cache on content hash so unchanged pages skipped re-parsing, cutting end-to-end runtime 34% and roughly halving peak memory.
- Built web automation tooling that extracted and normalized data from 10,000+ sources, handling pagination, rate limits, and inconsistent markup.

Projects

FRC Team 346 Competition Robot (Sole Software Lead, 4 Seasons) Jan 2024 – Dec 2024
Java, Real-Time Control, Sensor Fusion *GitHub*

- Wrote the full control stack in Java, including subsystem state machines, command scheduling, autonomous routines, and driver controls, running on a 50 Hz loop with a 20 ms per-cycle budget.
- Diagnosed dropped control cycles caused by blocking calls in the main loop; restructured the superstructure as a non-blocking state machine so per-cycle work stayed bounded.
- Implemented pose estimation fusing wheel odometry, gyroscope, and vision measurements for autonomous localization and path following.

Crucible (1st Place, Claude Buildathon) Apr 2026
Python, Multi-Agent Systems

- Built a deliberation tool that runs competing model agents against each other and surfaces their disagreements before producing a recommendation; placed 1st of 50+ teams.

Skills

Languages: C++, C, Python, Java, Bash, TypeScript
Systems & Robotics: state machines, sensor fusion, concurrency, memory debugging, performance profiling
Tools: Linux, GDB, Valgrind, Git, Make/CMake, Docker, PostgreSQL, SQLite, AWS, Google Cloud
ML: PyTorch, scikit-learn, LLM evaluation, RAG, multi-agent pipelines